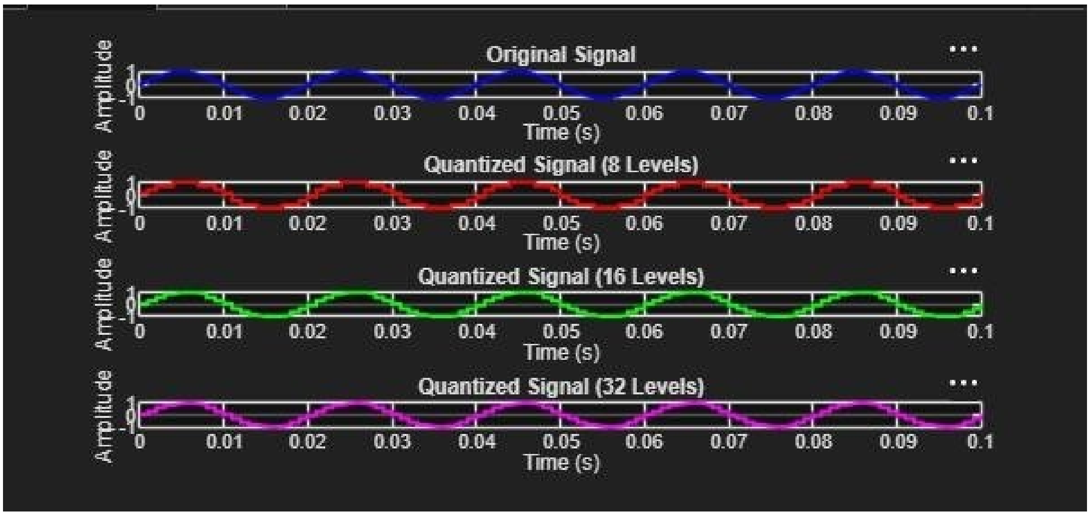
EXPT NO;6	Finite Word length Effects
DATE	6.1 Uniform Quantization and Quantization Error Analysis of a Discrete-Time Sinusoidal Signal Using MATLAB

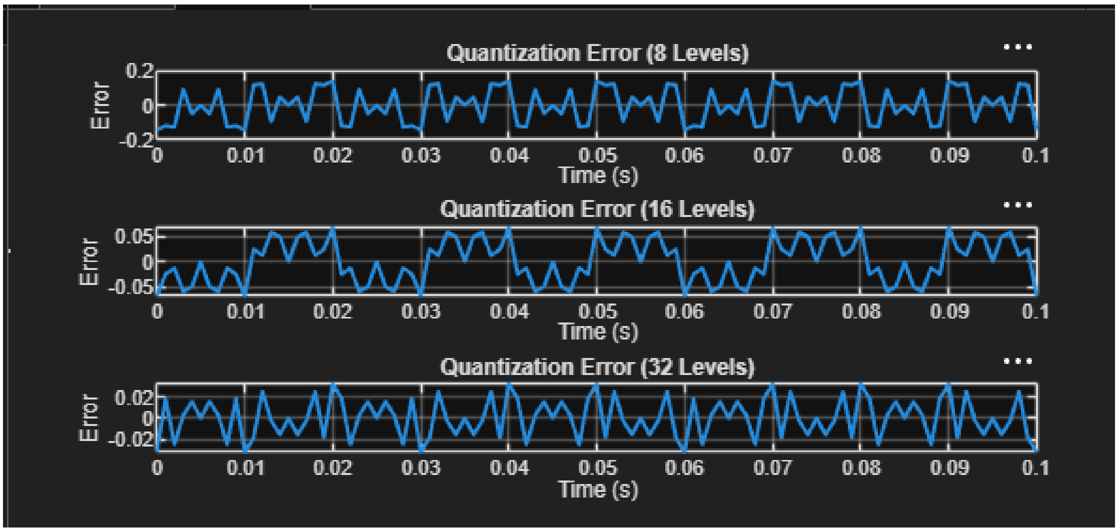
Program :

```
clc;
clear;
close all;
%% Signal
fs = 1000; f = 50;
t = 0:1/fs:0.1;
x = sin(2*pi*f*t);
%% Quantization Levels
L = [8 16 32];
colors = {'r','g','m'};
figure;
for i = 1:length(L)
    % Quantization
    xq{i} = round((x+1)(L(i)-1)/2)(2/(L(i)-1)) - 1;
    % Error & MSE
    e{i} = x - xq{i};
    mse(i) = mean(e{i}.^2);
    % Plot Quantized Signals
    subplot(4,1,i+1);
    stairs(t,xq{i},colors{i},'LineWidth',1.2);
    title(['Quantized Signal (' num2str(L(i)) ' Levels)']);
    grid on;
end
% Original Signal
subplot(4,1,1);
plot(t,x,'LineWidth',1.5);
title('Original Signal'); grid on;
% Display MSE
fprintf('MSE for 8 Levels  = %f\n', mse(1));
fprintf('MSE for 16 Levels = %f\n', mse(2));
fprintf('MSE for 32 Levels = %f\n', mse(3));
%% Error Plots
figure;
for i = 1:3
    subplot(3,1,i);
    plot(t,e{i});
    title(['Error (' num2str(L(i)) ' Levels)']);
    grid on;
end
```

Output



Original Signal
Quantized Signal (8 Levels)
Quantized Signal (16 Levels)
Quantized Signal (32 Levels)
Amplitude
Time (s)
1
0
-1
0
0.01
0.02
0.03
0.04
0.05
0.06
0.07
0.08
0.09
0.1



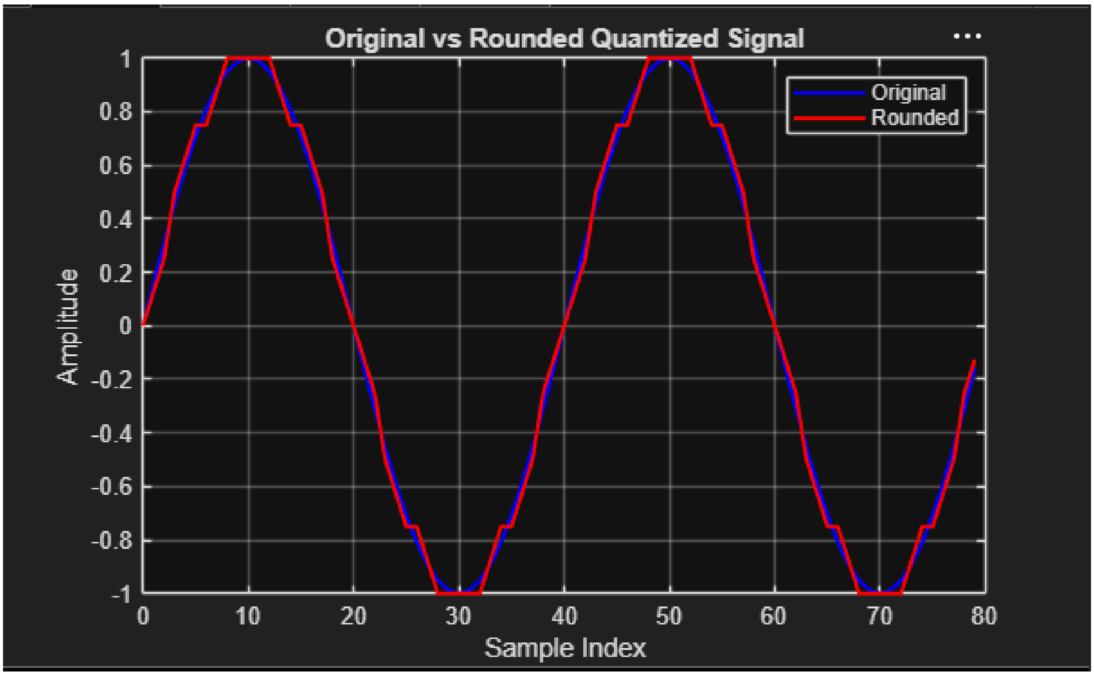
Expt 6	Finite Word length Effects
Date:	6.1 Uniform Quantization and Quantization Error Analysis of a Discrete-Time Sinusoidal Signal Using MATLAB

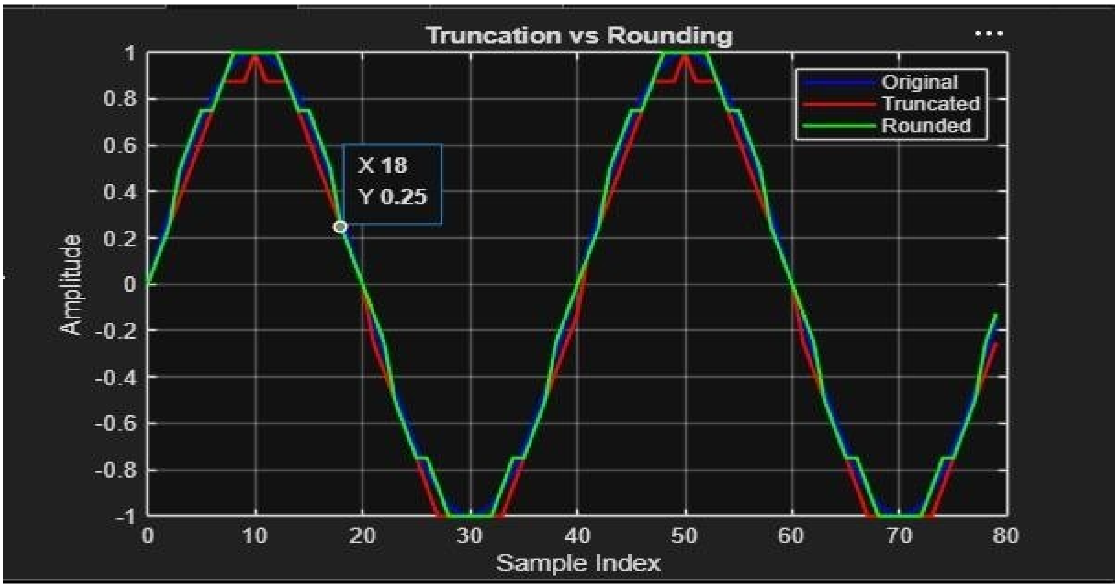
PROGRAM

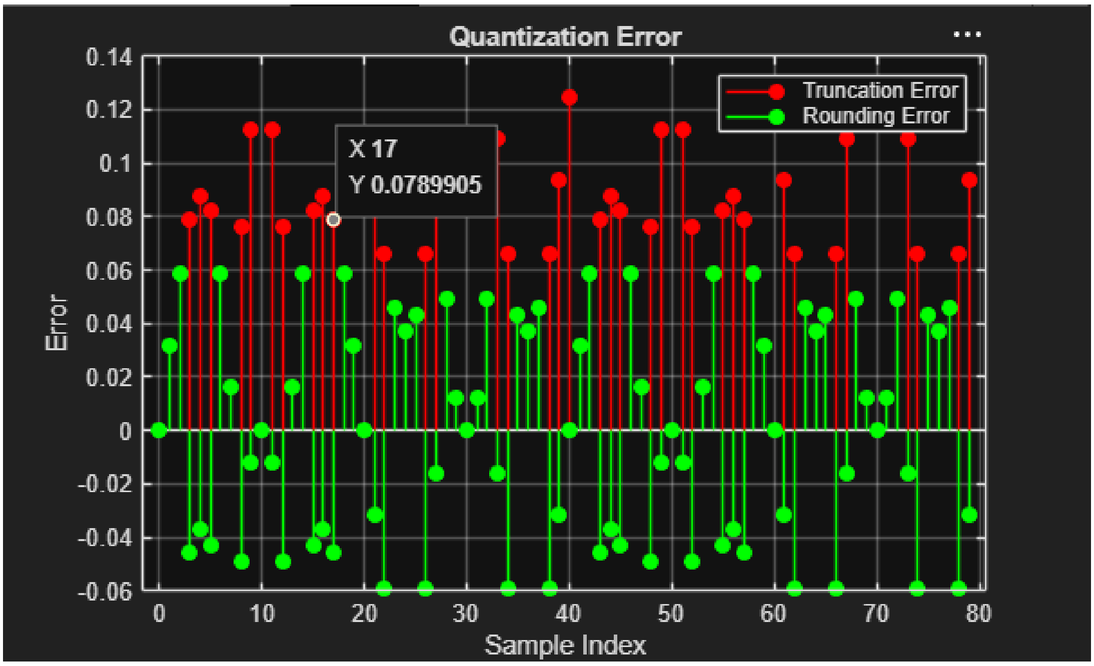
```
clc; clear; close all;
%% Generate Original Signal n = 0:79; % Sample Index
x = sin(2*pi*n/40); % Input Sinusoidal Signal

%% Quantization Factor
q = 8; % Number of Quantization Steps
%% Truncation and Rounding
xt = floor(x*q)/q; % Truncated Signal xr = round(x*q)/q; % Rounded Signal
%% Quantization Errors
et = x - xt; % Truncation Error er = x - xr; % Rounding Error
%% Mean Square Error mse_t = mean(et.^2); mse_r = mean(er.^2);

%% Figure 1: Original vs Rounded Signal figure;
plot(n,x,'b','LineWidth',1.5); hold on; plot(n,xr,'r','LineWidth',1.5); grid
on;
xlabel('Sample Index'); ylabel('Amplitude');
title('Original vs Rounded Quantized Signal'); legend('Original','Rounded');
%% Figure 2: Truncation vs Rounding figure;

plot(n,x,'b','LineWidth',1.5); hold on; plot(n,xt,'r','LineWidth',1.5);
plot(n,xr,'g','LineWidth',1.5); grid on;
xlabel('Sample Index'); ylabel('Amplitude'); title('Truncation vs Rounding');
legend('Original','Truncated','Rounded');

%% Figure 3: Quantization Error figure; stem(n,et,'r','filled');
hold on; stem(n,er,'g','filled'); grid on;
xlabel('Sample Index'); ylabel('Error'); title('Quantization Error');
legend('Truncation Error','Rounding Error');

%% Figure 4: MSE Comparison figure;
```

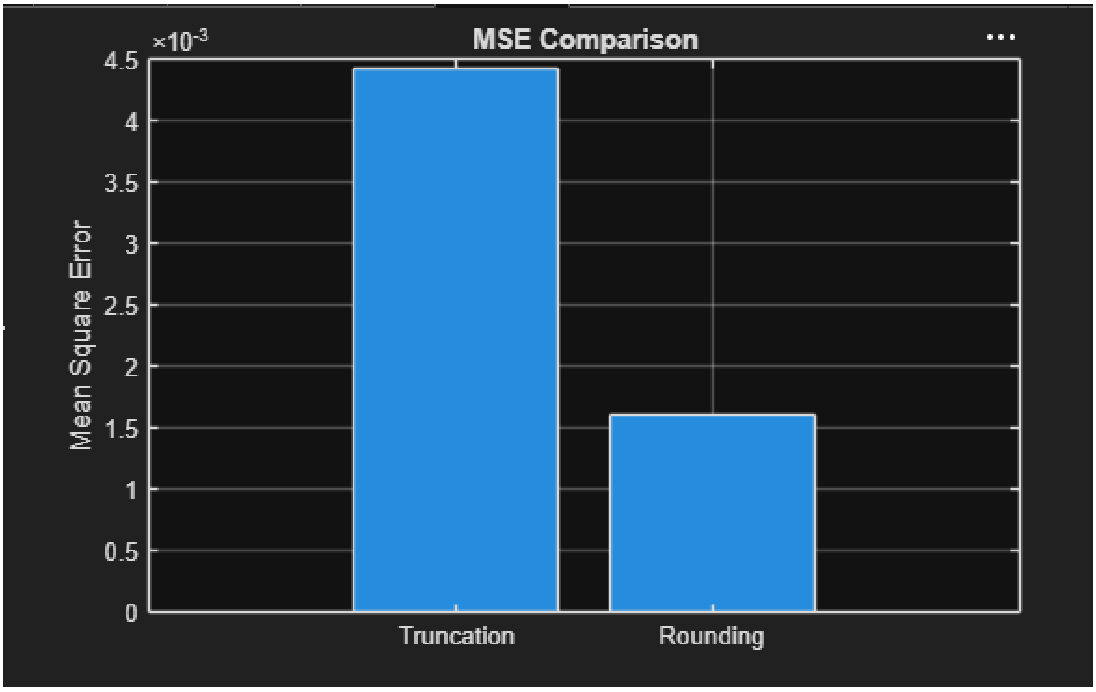
```
bar([mse_t mse_r]); set(gca,'XTickLabel',{'Truncation','Rounding'});
ylabel('Mean Square Error');
title('MSE Comparison'); grid on;

%% Display Results
fprintf('Mean Square Error (Truncation) = %.6f\n', mse_t); fprintf('Mean
Square Error (Rounding) = %.6f\n', mse_r);
```

Output:




Truncation vs Rounding
X 18
Y 0.25
Original
Truncated
Rounded
Amplitude
Sample Index
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1
0
10
20
30
40
50
60
70
80


Quantization Error
X 17
Y 0.0789905
Truncation Error
Rounding Error
Error
Sample Index
0.14
0.12
0.1
0.08
0.06
0.04
0.02
0
-0.02
-0.04
-0.06
0
10
20
30
40
50
60
70
80



MSE Comparison
×10-3
4.5
4
3.5
3
2.5
2
1.5
1
0.5
0
Mean Square Error
Truncation
Rounding